



PYTHON - SE - BACKEND

SOFTWARE ENGINEERING – MODULE 4

Database Management – SQL & PL/SQL

SESSION / ASSIGNMENT – 8

TASKS :

1. Write a PL/SQL block that declares two variables: one for a user's name (VARCHAR2) and one for their age (NUMBER). Assign values and print both using DBMS_OUTPUT.PUT_LINE.

→ CODE:

```
DECLARE
    user_name VARCHAR2(50);
    user_age NUMBER;
BEGIN
    user_name := 'Haaris';
    user_age := 22;

    DBMS_OUTPUT.PUT_LINE('User Name: ' || user_name);
    DBMS_OUTPUT.PUT_LINE('Age: ' || user_age);
END;
```

OUTPUT:

```
SQL> DECLARE
      user_name VARCHAR2(50);
      user_age NUMBER;
      BEGIN...
```

Show more...

```
User Name: Haaris
Age: 22
```

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.005

2. Create a PL/SQL block that uses an IF statement to check if a given order amount is above 500. If yes, print 'Eligible for free delivery', else print 'Delivery charges apply'.

→ CODE:

```
DECLARE
    order_amount NUMBER := 750;
BEGIN
    IF order_amount > 500 THEN
        DBMS_OUTPUT.PUT_LINE('Eligible for free delivery');
    ELSE
        DBMS_OUTPUT.PUT_LINE('Delivery charges apply');
    END IF;
END;
/
```

OUTPUT:

If order_amount > 500

```
SQL> DECLARE
    order_amount NUMBER := 750;
BEGIN
    IF order_amount > 500 THEN...
Show more...
```

Eligible for free delivery

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.005

If order_amount < 500

order_amount NUMBER := 250;

```
SQL> DECLARE
    order_amount NUMBER := 250;
BEGIN
    IF order_amount > 500 THEN...
Show more...
```

Delivery charges apply

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.004

3. Write a PL/SQL block that uses a simple LOOP to print the numbers 1 to 5 using DBMS_OUTPUT.PUT_LINE.

Hint: Use a counter variable and EXIT WHEN condition inside the loop.

➔ CODE:

```
DECLARE
    counter NUMBER := 1;
BEGIN
    LOOP
        DBMS_OUTPUT.PUT_LINE(counter);

        counter := counter + 1;

        EXIT WHEN counter > 5;
    END LOOP;
END;
/
```

OUTPUT:

```
SQL> DECLARE
      counter NUMBER := 1;
      BEGIN
      LOOP...
      Show more...
```

```
1
2
3
4
5
```

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.005

4. Declare a cursor in PL/SQL to select all product names from a table called PRODUCTS. Fetch each product name and print it using DBMS_OUTPUT.PUT_LINE.

Hint: Assume the PRODUCTS table has a column PRODUCT_NAME.

➔ CODE:

Product table

```
CREATE TABLE PRODUCTS (  
    PRODUCT_ID NUMBER PRIMARY KEY,  
    PRODUCT_NAME VARCHAR2(100)  
);
```

Inserting Data in Product table

```
INSERT INTO PRODUCTS (PRODUCT_ID, PRODUCT_NAME)  
VALUES (1, 'Laptop');
```

```
INSERT INTO PRODUCTS (PRODUCT_ID, PRODUCT_NAME)  
VALUES (2, 'Mobile Phone');
```

```
INSERT INTO PRODUCTS (PRODUCT_ID, PRODUCT_NAME)  
VALUES (3, 'Headphones');
```

```
COMMIT;
```

Declaring a cursor

```
DECLARE  
    CURSOR product_cursor IS  
        SELECT PRODUCT_NAME  
        FROM PRODUCTS;  
  
    product_name PRODUCTS.PRODUCT_NAME%TYPE;  
BEGIN  
    OPEN product_cursor;  
  
    LOOP  
        FETCH product_cursor INTO product_name;  
  
        EXIT WHEN product_cursor%NOTFOUND;  
  
        DBMS_OUTPUT.PUT_LINE('Product Name: ' || product_name);  
    END LOOP;
```

```
        CLOSE product_cursor;  
END;  
/
```

OUTPUT:

```
SQL> DECLARE  
        CURSOR product_cursor IS  
        SELECT PRODUCT_NAME  
        FROM PRODUCTS;...
```

Show more...

```
Product Name: Laptop  
Product Name: Mobile Phone  
Product Name: Headphones
```

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.045

5. Write a PL/SQL block that tries to divide two numbers entered as variables. If a division by zero occurs, catch the exception and print 'Cannot divide by zero'. Otherwise, print the result.

→ CODE:

```
DECLARE
    num1 NUMBER := 100;
    num2 NUMBER := 0;
    result NUMBER;
BEGIN
    result := num1 / num2;

    DBMS_OUTPUT.PUT_LINE('Result: ' || result);

EXCEPTION
    WHEN ZERO_DIVIDE THEN
        DBMS_OUTPUT.PUT_LINE('Cannot divide by zero');
END;
/
```

OUTPUT:

```
SQL> DECLARE
      num1 NUMBER := 100;
      num2 NUMBER := 0;
      result NUMBER;...
Show more...

Cannot divide by zero

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.005
```